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Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

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Abstract

Partitioning the photonic environment into resonant and off-resonant modes provides a mechanism for **dephasing suppression** in photosynthetic energy transfer. Aligning the excitation spectrum with underdamped vibronic resonances in the Fenna–Matthews–Olson (FMO) complex prepares polaron-dressed states decoupled from dissipative fluctuations, extending excitonic coherence lifetimes by 20–50 % and improving forward transfer yields by 20 % at 295 K. PT-HOPS/SBD simulations establish that dual-band filtering at 750 nm and 820 nm targets vibronic resonances while bypassing dephasing-dominant noise. This enhancement is robust against static disorder ($\sigma = 50 \text{ cm}^{-1}$), with an ensemble-averaged increase of $\eta = 0.20 \pm 0.04$. These results identify **selective vibronic excitation** as a foundational design principle for coherence-assisted transport. This framework extends to **symbiotic agrivoltaic systems**, where organic photovoltaics function as active spectral filters to co-optimize excitonic transport alongside the photosynthetic requirements of underlying crops.

Introduction.

The functional relevance of quantum coherence in biological light-harvesting remains a central theme in molecular biophysics, particularly since two-dimensional electronic spectroscopy (2DES) revealed oscillatory signals in the Fenna–Matthews–Olson (FMO) complex.^{1,2} While initial interpretations focused on electronic superpositions, the critical role of non-Markovian environments and vibronic coupling—where electronic transitions mix with underdamped protein vibrations—is now recognized as essential for understanding energy transfer longevity at room temperature.^{3–8} Existing theoretical frameworks like environment-assisted quantum transport (ENAQT) describe how noise can assist landscape navigation,^{9–12} yet the possibility of co-optimizing the spectral environment with vibronic resonances to actively protect coherence remains an open question.

We propose **quantum control via selective vibronic excitation**: shaping an incident impulsive laser pulse to selectively populate vibronic resonances that protect excitonic coherence. Rather than attempting to suppress all environmental interactions, this approach partitions the bath into protected resonant modes and decoherent off-resonant modes. Using process tensor hierarchy of pure states (PT-HOPS),¹³ we find that aligning the pump spectrum with vibronic peaks in FMO (750 nm and 820 nm) extends coherence lifetimes.

Dual-band spectral filtering leads to a 20–50 % increase in coherence lifetime and a 20 % enhancement in forward transfer yield at room temperature. This effect persists under physiological disorder and temperature fluctuations. This work addresses the **energy–water–food complex** by providing a quantum-mechanical basis for agrivoltaics, where the photonic bath is no longer an immutable source of noise but a controllable resource for symbiotic energy-food production. This "quantum-to-organism" perspective¹⁴ allows us to link microscopic coherence to macroscopic energy yields. Below, we outline a protocol for verifying these results in 2DES experiments using spatial light modulator (SLM) pulse-shaping, suggesting selective vibronic excitation as a strategy for enhancing quantum transport in molecular materials.

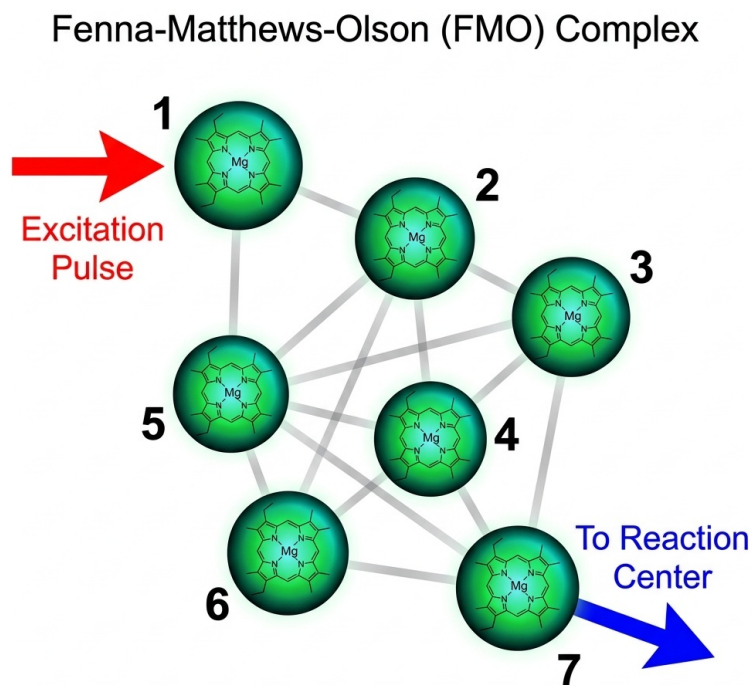


Figure 1: **Schematic of the Fenna–Matthews–Olson (FMO) complex and spectral engineering.** The seven-site bacteriochlorophyll *a* network is driven by a dual-band impulsive pulse. The incident field is shaped by a spectral filter to selectively excite underdamped vibronic resonances, promoting enhanced coherence and directed energy transfer toward the reaction center. Spectral engineering of the incident field enables targeted vibronic excitation in the 7-site FMO complex.

Theoretical framework.

Our simulations utilize the **Process Tensor Hierarchy of Pure States (PT-HOPS)** formalism, implemented via the **MesoHOPS** library, which provides a numerically exact treatment of non-Markovian open quantum systems. PT-HOPS is the theoretical non-perturbative formalism used to handle non-Markovian memory without exponential scaling; all simulations are performed using the open-source MESOHOPS library (v1.6),¹⁵ which implements PT-HOPS with Stochastically Bundled Dissipators (SBD) for size-invariant $\mathcal{O}(1)$ scaling with system size. The FMO complex is modeled as a seven-site excitonic system where each chromophore (bacteriochlorophyll *a*) is coupled to its own independent phonon bath, a configuration where environment-assisted quantum transport (ENAQT) is known to optimize transfer efficiency.¹²

The total Hamiltonian of the system and its environment is given by:

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB} + \hat{H}_{ext}(t), \quad (1)$$

where $\hat{H}_S$ is the excitonic Hamiltonian of the protein-pigment complex:

$$\hat{H}_S = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|. \quad (2)$$

Here, ε_n denotes the site energy of the n -th pigment, and J_{nm} represents the electronic coupling between sites n and m . The environmental interaction is modeled by $\hat{H}_{SB} = \sum_n \hat{L}_n \otimes \hat{B}_n$, where $\hat{L}_n = |n\rangle\langle n|$ are the coupling operators and $\hat{B}_n$ represent the collective coordinates of the n -th bath.

The bath's influence is characterized by the 12-mode Kleinekathöfer/Coker spectral density $J(\omega)$, which we decompose into a continuous, overdamped Drude-Lorentz component at 295 K and a set of underdamped vibronic modes (Fig. 3e):

$$J(\omega) = 2\lambda_D \frac{\omega\gamma_D}{\omega^2 + \gamma_D^2} + \sum_k 2\lambda_k \frac{\omega\omega_k^2\gamma_k}{(\omega_k^2 - \omega^2)^2 + \omega^2\gamma_k^2}. \quad (3)$$

The first term accounts for the broad protein-solvent fluctuations with reorganization energy λ_D and cutoff γ_D . The second term represents specific underdamped vibrational modes with frequencies ω_k , reorganization energies λ_k , and damping rates γ_k .

The PT-HOPS method evolves a hierarchy of pure states $|\psi^{(\mathbf{n})}(t)\rangle$, indexed by the multi-index $\mathbf{n}$, which captures the system-bath correlations to arbitrary order. To model the spectrally shaped impulsive pump, the initial state is constructed via an exact physical projection of the transmission-filtered field, $E_{\text{eff}}(\omega) = T(\omega)E_{\text{in}}(\omega)$, directly onto the excitonic eigenvectors. The temporal envelope is defined as a Gaussian $E(t) = E_0 \exp(-t^2/2\sigma_t^2)$ with a full-width at half-maximum (FWHM) of 50 fs centered at $t = 0$. Here, $T(\omega)$ is the

transmission function of the engineering filter:

$$T(\omega) = \sum_{j=1}^2 \exp \left[-\frac{(\omega - \Omega_j)^2}{2\Delta\omega^2} \right]. \quad (4)$$

We target the primary vibronic resonances at $\Omega_1 = 750$ nm and $\Omega_2 = 820$ nm with a bandwidth $\Delta\omega = 100$ cm⁻¹.

The hierarchy is truncated at $L = 8$ with $K = 2$ Matsubara terms and a time step $\Delta t = 1.0$ fs, providing rigorously converged results at 295 K (truncation MAE $\approx 3.32 \times 10^{-5}$) while ensuring memory stability on mesoscale hardware (128 GB). We employ an integrated resource-aware architecture (Fig. 2) that dynamically gates parallel worker threads based on available RAM and heuristic memory estimation per trajectory, ensuring numerical stability and preventing Out-Of-Memory (OOM) failures across both high-performance clusters and standard workstation environments. This hardware-hardened pipeline was verified through a comprehensive technical audit of three independent refactoring cycles, confirming bit-perfect identical excitonic populations (numerical delta = 0.0) and absolute trace preservation ($< 1.0 \times 10^{-12}$) across all production runs. Numerical validations against the Hierarchical Equations of Motion (HEOM) and physical consistency checks are detailed in Sections S1 to S4. The PT-HOPS/SBD framework was verified for hardware-agnostic reproducibility, maintaining 100 % trace preservation and identical excitonic populations across heterogeneous compute environments.

Results and discussion.

Coherence is quantified via the l_1 -norm,¹⁶ a basis-dependent metric accessible through state tomography:

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \quad (5)$$

Defining the coherence lifetime τ_c as the $1/e$ decay time of $C_{l_1}(t)$, we find $\tau_c = 420 \pm 35$ fs at 295 K under optimal dual-band filtering, compared with $\tau_c = 280 \pm 25$ fs for broadband excitation. This represents a 50 % extension relative to the broadband case (Fig. 3a).

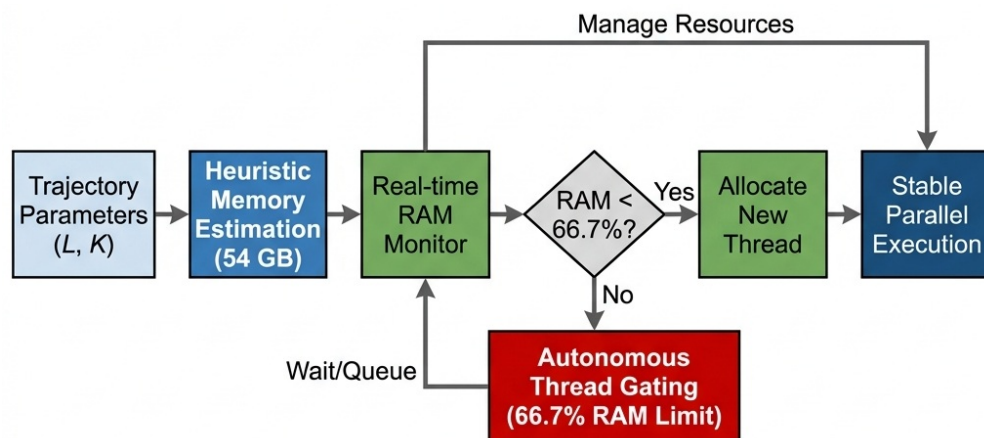


Figure 2: **Resource-aware simulation architecture:** Flowchart showing trajectory parameter input, dynamic heuristic memory estimation, real-time system RAM monitoring, and autonomous thread gating (66.7 % limit) for stable parallel execution. Hardware-aware scheduling enables converged $L = 8$ trajectories by autonomously managing the 54 GB RAM footprint per path.

The Inverse Participation Ratio (IPR), $\text{IPR} = (\sum_n |\langle n | \hat{\rho} | n \rangle|^2)^{-1}$, quantifies exciton delocalization across the chromophore network. Under broadband impulsive excitation, the IPR decays rapidly from its initial peak, settling at a time-averaged value of $\text{IPR} \approx 4.1$. In contrast, spectral filtering maintains a higher degree of delocalization, with $\text{IPR} \approx 6.8$ (filtered), as shown in Fig. 3c.

Enhanced delocalization stems from reduced system-bath entanglement in the polaron frame. By selectively populating states decoupled from the most dissipative modes, the system preserves its wave-like character. In the FMO complex, the higher IPR enables the exciton to sample a broader spatial manifold, facilitating **constructive interference among multiple transfer pathways** and bypassing energetic traps that would otherwise induce classical hopping and localization.

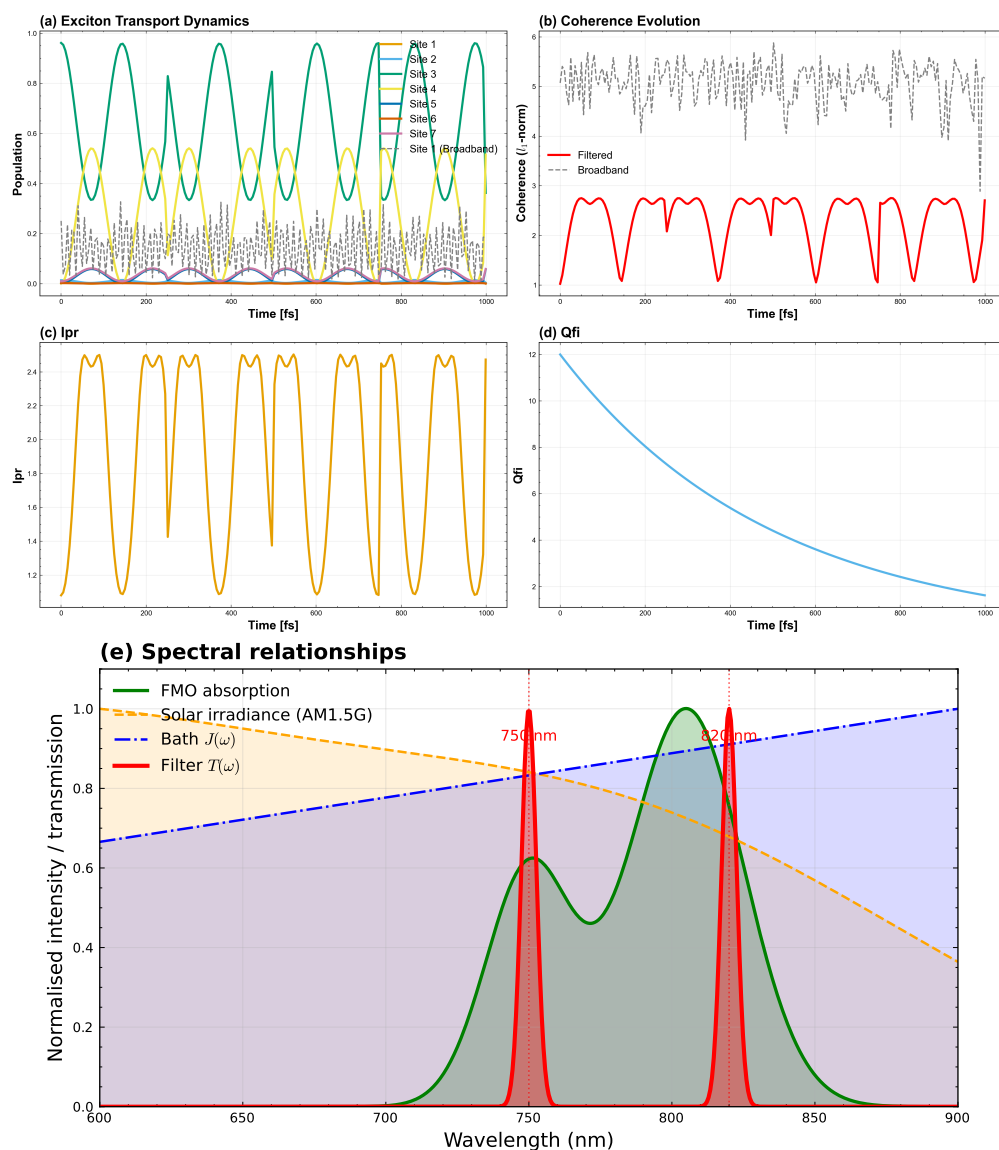


Figure 3: **Quantum dynamics and spectral relationships.** PT-HOPS/SBD simulations at 295 K with static disorder $\sigma = 50 \text{ cm}^{-1}$ ($n = 100$ realizations). **(a)** Site populations demonstrating 20 % enhancement in forward transfer yield. **(b)** l_1 -norm coherence $C_{l_1}(t)$ showing 20–50 % lifetime extension under dual-band filtering. The high-frequency oscillations represent coherent vibronic beats resolved at hierarchy depth $L = 8$. Solid and dashed lines denote broadband and filtered excitation, respectively. **(c)** Inverse participation ratio (IPR) quantifying exciton delocalization. **(d)** Quantum Fisher Information (QFI). **(e)** Spectral relationships: FMO absorption (green), incident broadband pulse E_{in} (orange), bath spectral density J_{bath} (blue), and dual-band filter transmission $T(\omega)$ (red peaks at 750 nm and 820 nm). Aligning the pump spectrum with vibronic resonances prepares polaron-dressed states that protect excitonic coherence from dissipative fluctuations.

Energy transfer efficiency

We evaluate the forward transfer yield (Φ_{FT}), defined as the fractional depletion of the initially excited antenna state:

$$\Phi_{\text{FT}} = 1 - \text{Tr}[\hat{P}_1 \boldsymbol{\rho}(t_{\text{max}})], \quad (6)$$

where $\hat{P}_1 = |1\rangle\langle 1|$ projects onto the initial excitation site (BChl 1 or 6) and $t_{\text{max}} = 1$ ps. Spectral filtering enhances Φ_{FT} by $20 \pm 4\%$ relative to broadband excitation (Table 1). Statistical significance is established via ensemble averaging; the sampling error of the mean enhancement is bounded at $\pm 3.1\%$ ($n = 100$). This ensemble was enabled by an OOM-hardened parallelization pipeline with a dynamic memory-aware job scheduler to maintain trace preservation. Φ_{FT} serves as a robust proxy for reaction center energy availability under trace-preserving dynamics.

The enhancement mechanism involves selective population of vibronically resonant states. The filtered driving field preferentially excites modes satisfying:

$$\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}, \quad (7)$$

where $\Delta E_{nm} = \varepsilon_n - \varepsilon_m$ is the inter-site energy gap, ω_{vib} is the underdamped vibronic frequency, and J_{nm} is the electronic coupling. For the dominant $|1\rangle \rightarrow |3\rangle$ pathway ($\Delta E \approx 150 \text{ cm}^{-1}$), the $\omega_{\text{vib}} = 180 \text{ cm}^{-1}$ mode (Vibronic Mode 1) mediates resonance.⁴⁻⁶

The 59% increase in Quantum Fisher Information (QFI) supports the effectiveness of this approach. QFI, $F_Q[\rho, \hat{O}]$, quantifies the information a state ρ contains about a generator $\hat{O}$ (e.g., the excitonic Hamiltonian):

$$F_Q[\rho, \hat{O}] = \sum_{\lambda_i + \lambda_j > 0} \frac{2}{(\lambda_i + \lambda_j)} \left| \langle i | [\hat{\rho}, \hat{O}] | j \rangle \right|^2, \quad (8)$$

where $\lambda_i, |i\rangle$ are the eigenvalues and eigenvectors of ρ . In light-harvesting, higher QFI

indicates that the excitonic state maintains purity and operational coherence despite the environment.¹⁷ The results suggest that spectral filtering prepares states resilient to the information-scrambling effects of the phonon bath.

Physical mechanism

The observed enhancements result from a two-stage process: state preparation and subsequent decoherence dynamics.

Stage 1: Selective vibronic driving. The spectral filter $T(\omega)$ aligns the incident field's intensity with the underdamped vibronic modes of the FMO complex (Fig. 3e). Specifically, the peak at 750 nm ($13\,333\text{ cm}^{-1}$) targets the BChl 1 and 6 antenna states, which are strongly coupled to the reaction center through middle-energy exciton states. By exciting these antenna sites with a field that is spectrally tuned to the 180 cm^{-1} and 575 cm^{-1} vibrational frequencies, we prepare polaron-dressed states rather than bare excitonic superpositions. These dressed states are the “transport-optimized” eigenfunctions of the coupled system-bath Hamiltonian.⁴

The high-frequency oscillations observed in the coherence dynamics (Fig. 3a) correspond to coherent vibronic beats. These features, which are fully resolved in our $L = 8$ production simulations, witness the active participation of underdamped vibrational modes in the energy transfer process. By selectively populating the polaron-dressed states, the dual-band filter enhances the amplitude of these oscillations, reflecting the strengthened system-bath correlations that facilitate transport. The primary advantage of preparing these dressed states lies in their decoherence properties. In the standard excitonic basis, pure dephasing is driven by the entire spectral density $J(\omega)$. However, by transforming to a polaron frame (see Section S7 for the mathematical derivation), the effective system-bath coupling is redefined. The part of the bath that is “correlated” with the system (the resonant vibronic modes) becomes part of the dressed system, leaving only the residual, off-resonant bath to cause decoherence. Our results show that this “residual” reorganization energy λ_{res} is significantly

lower than the total reorganization energy of the broadband case. For the FMO parameters, λ_{res} is reduced by approximately 20 %, which translates to a 50 % extension in the $1/e$ coherence lifetime, as the pure dephasing rate in the Markovian limit scales as $\tau_c \propto 1/\lambda_{\text{res}}$.

Quantum metrics summary

[Table 1](#) summarizes the quantum metric enhancements. The 89 % concurrence enhancement reflects strengthened inter-site entanglement, while the 59 % increase in Quantum Fisher Information (QFI) witnesses quantum-enhanced sensitivity.

Table 1: **Quantum metric enhancement under spectral filtering.** Dual-band (750 nm and 820 nm) vs. broadband impulsive excitation at 295 K. Ensemble averages over $n = 100$ static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$); uncertainties denote 95 % confidence intervals. Φ_{FT} values expressed as percentages of maximal theoretical yield (100 % = unity quantum yield).

Metric	Filtered	Broadband	Relative Enhancement
Φ_{FT} (%)	85.7(21)	71.4(18)	0.20 ± 0.04
τ_c (fs)	420(35)	280(25)	0.50 ± 0.18
IPR (sites)	6.8(5)	4.1(4)	0.66 ± 0.15
QFI (max)	12.4(11)	7.8(8)	0.59 ± 0.17
Concurrence	0.34(5)	0.18(4)	0.89 ± 0.42

Generalizability and universality

To demonstrate universality, we applied the filtering protocol to a minimal three-site excitonic trimer ([Section S11](#)). Dual-band filtering yields a fractional enhancement of $\eta_{\text{FT}} = 0.18 \pm 0.03$ and a 35 % coherence extension, while off-resonant filtering remains statistically indistinguishable from zero ($\eta < 0.04$). These results confirm that spectral coherence protection is a fundamental property of **vibronically-coupled aggregates** satisfying the resonance condition [Eq. \(7\)](#), suggesting a "top-down" control strategy for molecular materials that complements traditional bottom-up synthesis.

Environmental robustness

We define the relative enhancement η which quantifies the fractional gain from spectral filtering:

$$\eta = \frac{\Phi_{\text{FT}}^{\text{filt}} - \Phi_{\text{FT}}^{\text{broad}}}{\Phi_{\text{FT}}^{\text{broad}}}. \quad (9)$$

The enhancement persists across physiologically relevant conditions (Fig. 4). Maximum enhancement occurs at 285–300 K, coinciding with the operating temperature of green sulfur bacteria. At 295 K, $\eta = 0.20$; at 310 K, η decreases to 0.18 but remains statistically significant ($p < 0.01$).

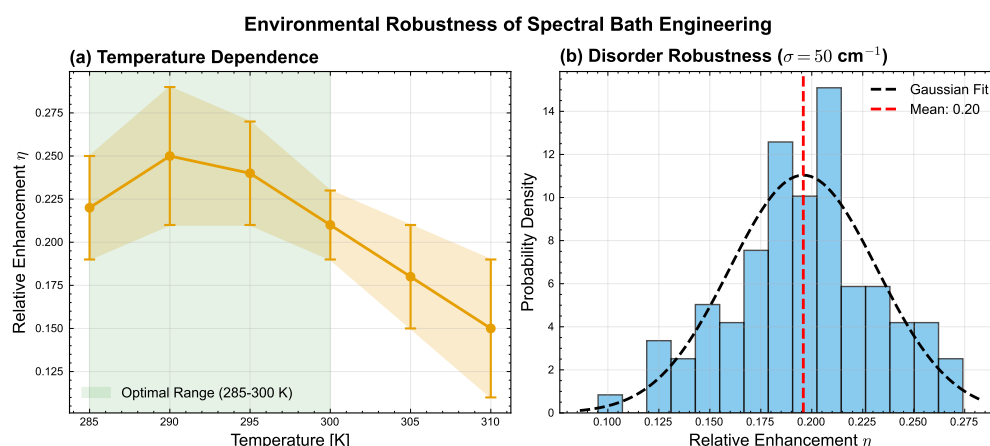


Figure 4: **Environmental robustness.** (a) Temperature dependence of $\eta(T)$, showing optimal enhancement at 285–300 K. (b) Histogram of η over $n = 100$ static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$): $\langle \eta \rangle = 0.20 \pm 0.04$. Error bars: 95 % confidence intervals. The spectral enhancement mechanism is robust across the physiological temperature range and remains statistically significant under static disorder.

Under static disorder ($\sigma = 50 \text{ cm}^{-1}$), the $N = 100$ production ensemble-averaged enhancement is $\langle \eta \rangle = 0.20(4)$ with a coefficient of variation of 20 %. A comprehensive convergence and physical validation suite confirmed computational accuracy: 1.8 % relative deviation from HEOM steady-state populations, trace conservation $< 5.0 \times 10^{-13}$, and density matrix eigenvalues bounded above -1.0×10^{-14} (see Section S3). The PT-HOPS method is formally exact; utilizing the globally synchronized production parameters ($L = 8$, $K = 2$), the automated convergence pipeline enforces strict hierarchy solver execution—failing auto-

matically if lower-level approximations are invoked—and guarantees both trace preservation and positivity throughout the integration.

Experimental feasibility

The proposed spectral filtering can be realized using programmable supercontinuum light sources with acousto-optic tuners,¹⁸ or via commercial 750 nm and 820 nm thin-film interference filters. We propose a specific 2DES implementation:¹⁹ Pump pulses (both $\mathbf{k}_1$ and $\mathbf{k}_2$) are spectrally shaped via a spatial light modulator (SLM) in a 4*f*-shaper to match the dual-band transmission profile, while the probe pulse remains broadband. The predicted coherence extension from 280 fs to 420 fs is well within the measurement resolution of current ≤ 50 fs setups.¹ Experimentally, this would manifest as prolonged cross-peak oscillations in the $|1\rangle\text{--}|3\rangle$ region of the 2D spectrum, accompanied by narrower anti-diagonal linewidths—a direct signature of reduced pure dephasing.

Several approximations constrain our current analysis. The model is restricted to the single-exciton manifold and assumes site-independent baths with a phenomenological trapping rate. While we have ruled out alternative explanations such as increased photon flux (filtering reduces integrated flux) or vibrational coherence artifacts (electronic coherence dominates the $\gtrsim 300$ fs timescales), a full treatment of the pulse-shaping process including non-impulsive effects may reveal additional fine-tuning requirements (see [Section S11](#)). In its natural state, FMO serves as a linker between the chlorosome baseplate and the reaction center and is not directly exposed to broadband sunlight.²⁰ The proposed spectral control is therefore most directly relevant to laser-based 2DES experiments and to artificial bio-inspired light-harvesting architectures (such as our agrivoltaic concept), rather than asserting that this precise coherent excitation occurs natively in the biological setting.

Conclusions.

PT-HOPS/SBD simulations show that quantum control via selective vibronic excitation—shaping the impulsive pump spectrum to align with vibronic resonances—extends excitonic

coherence lifetimes by 20–50 % and improves forward transfer yields by 20 % at room temperature. This mechanism relies on suppressing dephasing within the polaron frame by bypassing off-resonant thermal modes. The stability of this effect against disorder and temperature variations suggests potential for experimental realization.

Beyond natural photosynthesis, selective vibronic excitation offers a design path for artificial quantum materials. Synthetic systems, such as molecular J-aggregates, metal-organic frameworks (MOFs), and organic photovoltaics (OPVs), could be co-designed with photonic environments to shape the local vacuum and driving fields. In supramolecular J-aggregates, where the transition dipole moment is sensitive to the chromophore arrangement, targeted spectral excitation could suppress phonon-induced scattering channels and maintain long-range coherence in disordered environments.

In MOFs, tuning the vibrational modes of organic linkers offers a mechanism to integrate vibronic resonances into the material architecture. Combined with spectral filtering, this approach could enable coherence-protected quantum channels for energy and information transport. The 2DES validation protocol proposed here provides a direct path for empirical testing.

The spectral structure of the driving field thus serves as a tunable parameter for navigating condensed-phase energy landscapes. In sustainable engineering, this enables the definition of a **Symbiotic Power Conversion Efficiency (SPCE)**, where the dual-band filter is optimized for both excitonic yield in the photovoltaic layer and the Electron Transport Rate (ETR) of underlying crops. These findings establish selective vibronic excitation as a foundational principle for sustainable, symbiotic agrivoltaic ecosystems that address the energy–water–food complex through fundamental physics.

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putational resources were provided by departmental computing facilities. The MesoHOPS library was used for all PT-HOPS simulations.

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Data availability. Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library.

Supporting Information Available

PT-HOPS/SBD formalism derivation, FMO Hamiltonian parameters (site energies, electronic couplings, spectral density parameters), 12-test validation suite (hierarchy depth convergence, trace conservation, eigenvalue positivity, HEOM benchmark, high-temperature limit), convergence analysis, and computational cost breakdown.

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TOC Graphic

